

Square and Cube Roots

Objective: calculate and estimate square and cube roots.

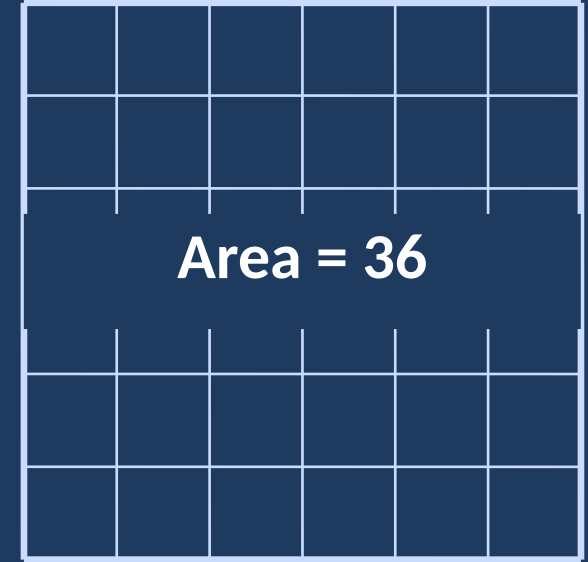
Success criteria: I can ...

SC1 find $\sqrt{\quad}$ of a square number and $\sqrt[3]{\quad}$ of a cube number

SC2 explain that a root undoes a power, and check my answer

SC3 say which two whole numbers a root lies between

SC4 estimate a root to 1 decimal place



side = ?

Starter: 3 minutes, no calculator

STARTER • 00-04

S1 $8^2 =$

S2 $11^2 =$

S3 $3^3 =$

S4 $10^3 =$

S5 $4^3 =$

S6 Which are square numbers?
20 25 36 48 81

Worksheet page 1 • Work silently

Starter: answers

STARTER • 00-04

s1 $8^2 = 8 \times 8 = 64$

s2 $11^2 = 121$

s3 $3^3 = 3 \times 3 \times 3 = 27$

s4 $10^3 = 1000$

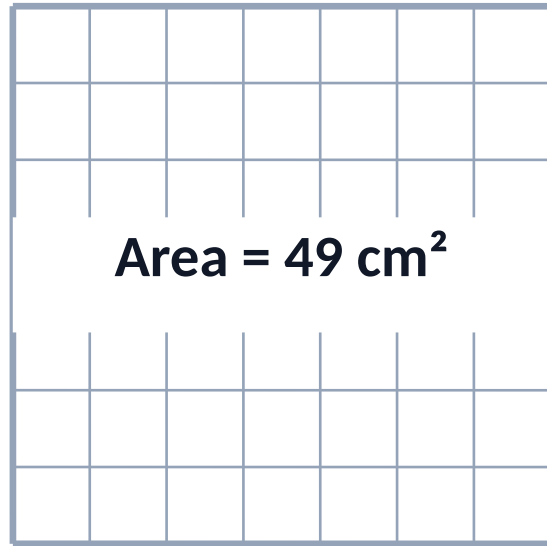
s5 $4^3 = 4 \times 4 \times 4 = 64$

s6 $25, 36, 81$
 $(5^2, 6^2, 9^2)$

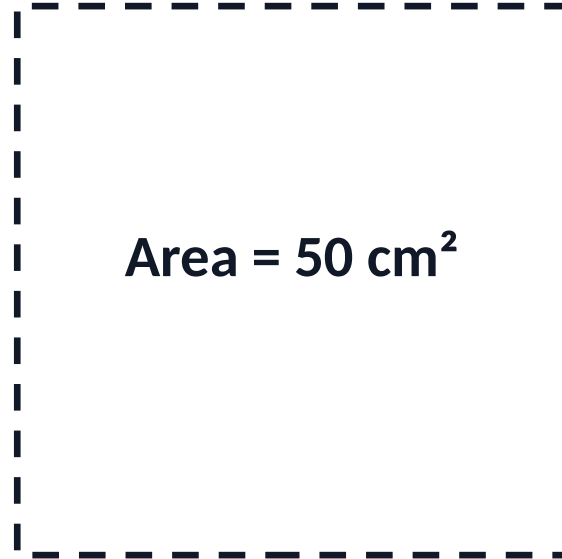
Tick or correct each answer. Hands up: 6/6? 5/6? 4 or fewer?

From your preparation

PREP CHECK • 04-07



side = ?



side = ?

1 Side of the 49 cm² square?

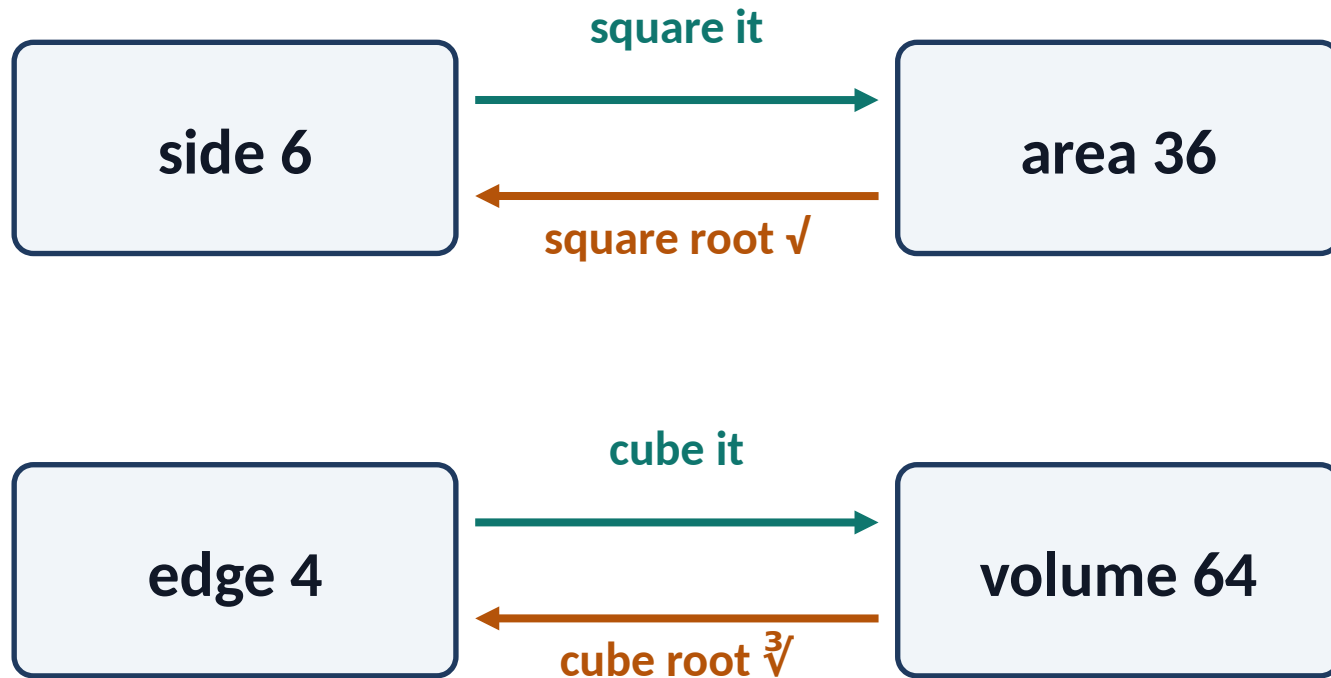
2 Area 50 cm²: between which two whole numbers?

3 Omar says $5^2 = 10$. Agree?

Answer on your mini-whiteboard

A root undoes a power

I DO • 07-08



Vocabulary

$\sqrt{36} = 6$ because $6^2 = 36$

$\sqrt[3]{64} = 4$ because $4^3 = 64$

$\sqrt{\quad}$ means the positive root

inverse: an operation that undoes another

estimate: a close value, found by reasoning

consecutive: next in order, e.g. 6, 7

Example 1: exact roots

I DO • 08-10

a $\sqrt{81}$

b $\sqrt[3]{125}$

c $\sqrt[3]{-27}$

Predict (c) on your fingers:
1 = positive 2 = negative 3 = impossible

Example 1: exact roots (solution)

I DO • 08-10

(a) $\sqrt{81}$

Which number $\times$ itself = 81?

$$9 \times 9 = 81$$

$$\sqrt{81} = 9$$

Check: $9^2 = 81$ ✓

(b) $\sqrt[3]{125}$

Which number cubed = 125?

$$5 \times 5 \times 5 = 25 \times 5 = 125$$

$$\sqrt[3]{125} = 5$$

Check: $5^3 = 125$ ✓

(c) $\sqrt[3]{(-27)}$

neg $\times$ neg $\times$ neg = neg

$$(-3) \times (-3) \times (-3) = 9 \times (-3) = -27$$

$$\sqrt[3]{(-27)} = -3$$

Negative numbers have cube roots

Example 2: estimate $\sqrt{40}$ (steps 1-2)

I DO • 10-13

1 Square numbers either side: $36 < 40 < 49$, so $6 < \sqrt{40} < 7$



2 Closer to? $40 - 36 = 4$, $49 - 40 = 9$. So $\sqrt{40}$ is closer to 6.

Example 2: estimate $\sqrt{40}$ (steps 3–4)

I DO • 10–13

3

Trap it:

$$6.3^2 = 39.69 \text{ (too small)}$$

$$6.4^2 = 40.96 \text{ (too big)}$$

4

Closer one:

$$40 - 39.69 = 0.31$$

$$40.96 - 40 = 0.96$$



$\sqrt{40} \approx 6.3$ (1 d.p.) Calculator check: 6.324... ✓

Example 3: estimate $\sqrt[3]{50}$

I DO • 13-15

1 Cube numbers either side: $27 < 50 < 64$, so $3 < \sqrt[3]{50} < 4$

2 Closer to? $50 - 27 = 23$, $64 - 50 = 14$. Closer to 4.

3 Trap it: $3.6^3 = 46.656$ (too small), $3.7^3 = 50.653$ (too big)

4 Closer one: 3.344 vs 0.653 . $\sqrt[3]{50} \approx 3.7$ (calculator $3.684...$ ✓)

Same 4 steps, but CUBE numbers: 1, 8, 27, 64, 125

Mini-whiteboards: Round 1

CHECK 1 • 15–19

1.1

$$\sqrt{9}$$

1.2

$$\sqrt{100}$$

1.3

$$\sqrt[3]{64}$$

1.4

A square has area 81 cm^2 . How long is a side?

1.5

True or false? $\sqrt{36} = 18$

Boards down → work alone → show in 3, 2, 1 → do not erase

Round 1: answers

CHECK 1 • 15–19

1.1

$$\sqrt{9} = 3$$

1.2

$$\sqrt{100} = 10$$

1.3

$$\sqrt[3]{64} = 4$$

1.4

$$\sqrt{81} = 9, \text{ so the side is } 9 \text{ cm}$$

1.5

$$\text{FALSE: } \sqrt{36} = 6 \\ (\text{check: } 18^2 = 324)$$

Every root answer can be checked: square it or cube it.

WE DO: W1 and W2

WE DO • 19-24

W1 Find $\sqrt{144}$ and $\sqrt[3]{1000}$

____ $\times$ ____ = 144, so $\sqrt{144} =$ ____ Check: ____² = 144

____ $\times$ ____ $\times$ ____ = 1000, so $\sqrt[3]{1000} =$ ____

W2 Estimate $\sqrt{70}$ to 1 decimal place

Square numbers either side: ____ < 70 < ____, so ____ < $\sqrt{70}$ < ____

Closer to ____ because _____

Trap it: ____² = ____ and ____² = ____ . So $\sqrt{70} \approx$ ____

Worksheet page 3 • W1 together, W2 in pairs

WE DO: W3 and W4

WE DO • 24–27

W3

Estimate $\sqrt[3]{20}$ to 1 decimal place.
First step: $8 < 20 < 27$

W4

You know $\sqrt{16} = 4$. Find $\sqrt{1600}$ and $\sqrt{0.16}$.
Check each by squaring.

W3: 90 seconds on your own • W4: no help

WE DO: solutions

WE DO • 19-27

W1

$$12 \times 12 = 144, \text{ so } \sqrt{144} = 12$$

$$10 \times 10 \times 10 = 1000, \text{ so } \sqrt[3]{1000} = 10$$

W2

$$64 < 70 < 81 \rightarrow \text{closer to } 8$$

$$8.3^2 = 68.89, 8.4^2 = 70.56 \rightarrow \sqrt{70} \approx 8.4$$

W3

$$8 < 20 < 27 \rightarrow \text{closer to } 3$$

$$2.7^3 = 19.683, 2.8^3 = 21.952 \rightarrow \sqrt[3]{20} \approx 2.7$$

W4

$$40^2 = 1600, \text{ so } \sqrt{1600} = 40$$

$$0.4^2 = 0.16, \text{ so } \sqrt{0.16} = 0.4 \text{ (not } 0.04)$$

Tick, or correct in a different colour

Mini-whiteboards: Round 2

CHECK 2 • 27–31

2.1

$\sqrt{20}$ is between which two whole numbers?

2.2

$\sqrt{45}$: between which two? Closer to which?

2.3

$\sqrt[3]{30}$ is between which two whole numbers?

2.4

Estimate $\sqrt{10}$ to 1 decimal place.

2.5

True or false?
" $\sqrt{65} \approx 8.5$ because it is between 8 and 9"

Boards down → work alone → show in 3, 2, 1 → do not erase

Round 2: answers

CHECK 2 • 27–31

2.1 $16 < 20 < 25$, so 4 and 5

2.2 6 and 7; closer to 7 (4 away vs 9 away)

2.3 $27 < 30 < 64$, so 3 and 4

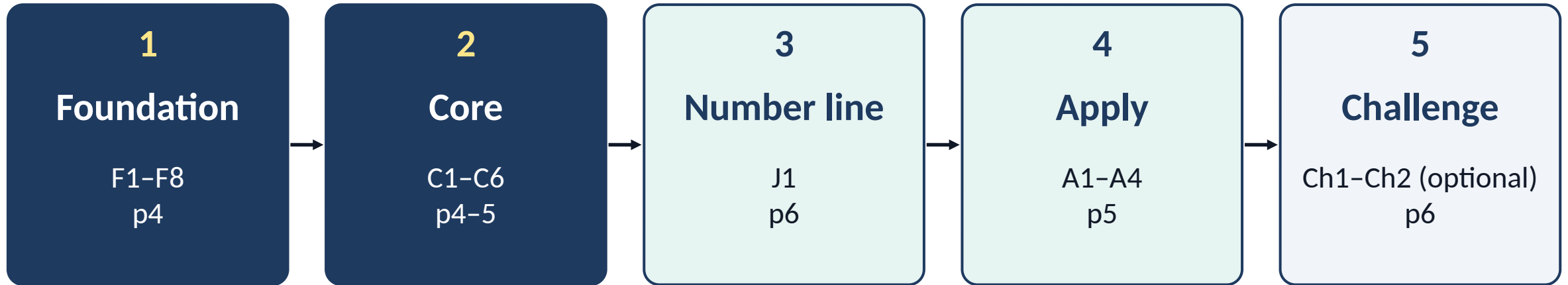
2.4 $3.1^2 = 9.61$, $3.2^2 = 10.24 \rightarrow \sqrt{10} \approx 3.2$

2.5 FALSE: 65 is just above 64, so $\sqrt{65}$ is just above 8. $8.1^2 = 65.61 \rightarrow \sqrt{65} \approx 8.1$

A root is NOT automatically halfway between the two whole numbers.

YOU DO: 10 minutes

YOU DO • 31-41



- Everyone: steps 1 and 2 (Foundation and Core).
- 5 minutes silent, then I will come round.
- Check every answer by squaring or cubing.
- Stuck? Use the Support Box at the top of page 4.
- Finish at 41:00.

Spot the Mistake

CHECK · 41–44

SM1 Ali: " $\sqrt{16} = 8$ because $16 \div 2 = 8$."

SM2 Bea: " $49 < 60 < 64$, so $\sqrt{60}$ is between 7 and 8. It must be in the middle, so $\sqrt{60} = 7.5$."

SM3 Cai: " $\sqrt[3]{-8}$ has no answer, because you cannot find a root of a negative number."

For each: circle the FIRST error · explain it · correct it (worksheet pages 6–7)

Spot the Mistake: corrections

CHECK • 41-44

SM1 Square root is not halving. $4 \times 4 = 16$, so $\sqrt{16} = 4$. (Check: $8^2 = 64$.)

SM2 $60 - 49 = 11$, $64 - 60 = 4$, so $\sqrt{60}$ is closer to 8.
 $7.7^2 = 59.29$, $7.8^2 = 60.84 \rightarrow \sqrt{60} \approx 7.7$. ($7.5^2 = 56.25$)

SM3 $(-2) \times (-2) \times (-2) = -8$, so $\sqrt[3]{-8} = -2$.
Only square roots of negatives have no real value.

Exit Ticket: 5 minutes, no calculator

EXIT • 44-49

X1

Find (a) $\sqrt{49}$ (b) $\sqrt[3]{27}$

X2

Between which two consecutive whole numbers is $\sqrt{85}$?

X3

Estimate $\sqrt{85}$ to 1 decimal place. Show a check.

X4

A cube has volume 8000 cm^3 . Find the edge length. Explain how you know.

Worksheet page 8 • on your own • hand in when finished

Recap and next step

1 Square or cube numbers either side

2 Which one is it closer to?

3 Trap it between decimals

4 Choose the closer one, then check

Homework

After-Lesson Practice (20–30 min)

Required: R1, Q1–Q10, Self-Test

Optional: Q11–Q13

Short on time: R1, Q1, Q2, Q5, Q6, Q8, Self-Test

Show 1–4 fingers: how confident are you with SC4 (estimating to 1 d.p.)?